EMA51005 - Discrete Mathematics
UNIT IV - Algebraic Systems
Final Exam Answer Key

This answer key follows the handwritten classwork flow: Closure → Associative → Identity → Inverse → Commutative whenever a group/abelian group is to be proved. Powers are typeset using normal superscripts.

Part A - Short Answers

1. **If $a * b = a + b + 2ab$ on real numbers, find the inverse element if it exists.**
   Answer: Identity is $e = 0$. Let a^{-1} be the inverse. Then

   $$a * a^{-1} = 0.$$

   Hence

   $$a + a^{-1} + 2aa^{-1} = 0,$$
   $$a^{-1}(1 + 2a) = -a.$$

   Therefore

   $$\boxed{a^{-1} = -\frac{a}{1+2a}}, \qquad a \neq -\frac{1}{2}.$$

2. **When is an algebraic system said to be a monoid?**
   Answer: A non-empty set G with a binary operation $*$ is called a monoid if it satisfies closure, associativity and has an identity element.

3. **Prove that every cyclic group is abelian.**
   Answer: Let $G = \langle a \rangle$. For $b, c \in G$, let $b = a^m$ and $c = a^n$. Then

   $$b * c = a^m * a^n = a^{m+n} = a^{n+m} = a^n * a^m = c * b.$$

   Hence G is abelian.

4. **Define a cyclic group with an example.**
   Answer: A group G is cyclic if there exists $a \in G$ such that every element of G is a power of a; that is $G = \langle a \rangle$. Example: $(\mathbb{Z}, +) = \langle 1 \rangle$.

5. **Find the identity for $a * b = a + b + 2$.**
   Answer: Let e be the identity. Then

   $$a * e = a \Rightarrow a + e + 2 = a.$$

   Therefore

   $$\boxed{e = -2}.$$

6. **Define a normal subgroup.**
   Answer: A subgroup H of G is normal if

   $$aH = Ha \quad \text{for every } a \in G.$$

   Equivalently, $a^{-1}Ha = H$ for every $a \in G$.

7. **Give an example of three distinct right cosets of a group G.**
   Answer: Take $G = \mathbb{Z}_6$ under addition and $H = \{0, 3\}$. Three distinct right cosets are

   $$H = \{0, 3\}, \quad H + 1 = \{1, 4\}, \quad H + 2 = \{2, 5\}.$$

8. **Prove that every subgroup of an abelian group is normal.**
   Answer: Let $H \leq G$ and G be abelian. For $a \in G$ and $h \in H$,

   $$a * h = h * a.$$

   Hence $aH = Ha$. Therefore H is normal in G.

9. **Define left coset and right coset.**
   Answer: If $H \leq G$ and $a \in G$, then

   $$aH = \{a * h : h \in H\}$$

   is the left coset, and

   $$Ha = \{h * a : h \in H\}$$

   is the right coset.

10. **What do you mean by code and decode?**
    Answer: Coding is converting a message into a prescribed coded form. Decoding is recovering the original message from the coded form using the corresponding rule or key.

Part B - 6 Mark Answers

1. **Prove that the identity element of a group is unique.**
Answer: Let $(G, *)$ be a group. Assume e_1 and e_2 are identity elements. For $a \in G$,

$$a * e_1 = e_1 * a = a, \qquad a * e_2 = e_2 * a = a.$$

Taking $a = e_1$ in the second relation,

$$e_1 * e_2 = e_1.$$

Taking $a = e_2$ in the first relation,

$$e_1 * e_2 = e_2.$$

Hence

$$\boxed{e_1 = e_2}.$$

Therefore the identity element is unique.

2. **If $(G, *)$ is abelian, show that $(a * b)^n = a^n * b^n$ for every positive integer n.**
Answer: Given G is abelian,

$$a * b = b * a.$$

We prove by mathematical induction.
For $n = 1$,

$$(a * b)^1 = a * b = a^1 * b^1.$$

Assume for $n = m$,

$$(a * b)^m = a^m * b^m.$$

For $n = m + 1$,

$$\begin{aligned}(a * b)^{m+1} &= (a * b)^m * (a * b)\\ &= (a^m * b^m) * (a * b)\\ &= a^m * (b^m * a) * b\\ &= a^m * (a * b^m) * b\\ &= (a^m * a) * (b^m * b)\\ &= a^{m+1} * b^{m+1}.\end{aligned}$$

Hence

$$\boxed{(a * b)^n = a^n * b^n}.$$

3. **Give examples of functions $\mathbb{N} \to \mathbb{N}$ which are (i) one-one but not onto, (ii) onto but not one-one, (iii) neither.**
Answer: **(i)** $f(n) = 2n$. If $f(a) = f(b)$, then $2a = 2b \Rightarrow a = b$, so it is one-one. Odd natural numbers have no pre-image, so it is not onto.
(ii) Define

$$f(2n-1) = n, \qquad f(2n) = n.$$

Every natural number has a pre-image, so it is onto. But $f(1) = f(2) = 1$, so it is not one-one.
(iii) $f(n) = 1$ for every n. Then many inputs have the same image and numbers $2, 3, \ldots$ have no pre-image. Hence it is neither one-one nor onto.

4. **Show that $S = \mathbb{Q} \times \mathbb{Q}$ is a semigroup for $(a, b) * (x, y) = (ax, ay + b)$.**
Answer: To prove a semigroup, prove closure and associativity.
(i) Closure: Let $(a, b), (x, y) \in S$. Then

$$(a, b) * (x, y) = (ax, ay + b).$$

Since $a, b, x, y \in \mathbb{Q}$, both ax and $ay + b$ are rational. Hence

$$(ax, ay + b) \in S.$$

(ii) Associative: Let $(a,b),(x,y),(p,q) \in S$.

$$\begin{aligned}[(a,b)*(x,y)]*(p,q) &= (ax, ay+b)*(p,q)\\ &= (axp, axq+ay+b).\end{aligned}$$

Also,

$$\begin{aligned}(a,b)*[(x,y)*(p,q)] &= (a,b)*(xp, xq+y)\\ &= (axp, a(xq+y)+b)\\ &= (axp, axq+ay+b).\end{aligned}$$

Hence associativity is satisfied. Therefore $(S,*)$ is a semigroup.

5. **Show that every group of order 3 is cyclic.**
Answer: Let
$$G = \{e, a, b\}$$
with $a \neq e$. By closure, $a^2 \in G$. If $a^2 = a$, cancellation gives $a = e$, impossible. If $a^2 = e$, then $a^{-1} = a$. Since $b \neq e, a$, this is impossible in a group of three elements because the inverse of the remaining nonidentity element would force a repetition. Hence
$$a^2 = b.$$
Then
$$a^3 = a^2 * a = b * a = e.$$
Therefore
$$G = \{e, a, a^2\} = \langle a \rangle.$$
Hence every group of order 3 is cyclic.

6. **Prove that G is abelian if and only if $(a*b)^2 = a^2 * b^2$.**
Answer: First, assume G is abelian.

$$\begin{aligned}(a*b)^2 &= (a*b)*(a*b)\\ &= a*(b*a)*b\\ &= a*(a*b)*b\\ &= (a*a)*(b*b)\\ &= a^2 * b^2.\end{aligned}$$

Conversely, assume
$$(a*b)^2 = a^2 * b^2.$$
Then
$$a*b*a*b = a*a*b*b.$$
By left cancellation of a,
$$b*a*b = a*b*b.$$
By right cancellation of b,
$$b*a = a*b.$$
Therefore
$$a*b = b*a.$$
Hence G is abelian.

7. **Show that $\mathbb{Q}$ is a monoid for $a * b = a + b + 2ab$.**
Answer: A monoid requires closure, associativity and identity.
(i) Closure: For $a, b \in \mathbb{Q}$,
$$a * b = a + b + 2ab \in \mathbb{Q}.$$
(ii) Associative:
$$\begin{aligned} a * (b * c) &= a + (b + c + 2bc) + 2a(b + c + 2bc) \\ &= a + b + c + 2ab + 2ac + 2bc + 4abc. \end{aligned}$$
Also,
$$\begin{aligned} (a * b) * c &= (a + b + 2ab) + c + 2(a + b + 2ab)c \\ &= a + b + c + 2ab + 2ac + 2bc + 4abc. \end{aligned}$$
Hence $a * (b * c) = (a * b) * c$.
(iii) Identity: Let e be identity.
$$a * e = a \Rightarrow a + e + 2ae = a.$$
Thus
$$e(1 + 2a) = 0.$$
Take $e = 0$. Then
$$a * 0 = 0 * a = a.$$
Therefore $e = 0$. Hence $(\mathbb{Q}, *)$ is a monoid.

8. **Show that the intersection of two subgroups of G is also a subgroup of G.**
Answer: Let $H_1, H_2 \leq G$. Since $e \in H_1$ and $e \in H_2$,
$$e \in H_1 \cap H_2,$$
so the intersection is non-empty. Let $a, b \in H_1 \cap H_2$. Then $a, b \in H_1$ and $a, b \in H_2$. Since both are subgroups,
$$a * b^{-1} \in H_1, \qquad a * b^{-1} \in H_2.$$
Hence
$$a * b^{-1} \in H_1 \cap H_2.$$
By the subgroup test,
$$\boxed{H_1 \cap H_2 \leq G}.$$

9. **Show that $\{1, 3, 7, 9\}$ is an abelian group under multiplication modulo 10.**
Answer: Let $G = \{1, 3, 7, 9\}$.
(i) Closure: The table is

$\times_{10}$	1	3	7	9
1	1	3	7	9
3	3	9	1	7
7	7	1	9	3
9	9	7	3	1

Every entry belongs to G.
(ii) Associative: Multiplication modulo 10 is associative because ordinary integer multiplication is associative.
(iii) Identity: $e = 1$ because $a \cdot 1 \equiv 1 \cdot a \equiv a \pmod{10}$.
(iv) Inverse:
$$1^{-1} = 1, \qquad 3^{-1} = 7, \qquad 7^{-1} = 3, \qquad 9^{-1} = 9.$$
(v) Commutative: Integer multiplication is commutative, hence multiplication modulo 10 is commutative.
Therefore G is an abelian group.

10. **Show that the intersection of two normal subgroups of G is also a normal subgroup of G.**
Answer: Let H and K be normal subgroups of G. First, by the subgroup test, $H \cap K$ is a subgroup of G.
Let $h \in H \cap K$ and $a \in G$. Since H is normal,

$$a^{-1} * h * a \in H.$$

Since K is normal,

$$a^{-1} * h * a \in K.$$

Therefore

$$a^{-1} * h * a \in H \cap K.$$

Hence $H \cap K$ is normal in G.

Part C - 10 Mark Answers

Classwork note: the printed questions for the first two group problems contain set-exclusion typos. The answers below use the handwritten classwork interpretation: $\mathbb{R} - \{-\frac{1}{2}\}$ *for* $a * b = a + b + 2ab$*, and* $\mathbb{R} - \{-1\}$ *for* $a * b = a + b + ab$*.*

1. **For** $G = \mathbb{R} - \{-\frac{1}{2}\}$ **and** $a * b = a + b + 2ab$**, show that** $(G, *)$ **is an abelian group.**
Answer: We prove the five conditions.
(i) Closure: Let $a, b \in G$. Then
$$a * b = a + b + 2ab.$$
If $a * b = -\frac{1}{2}$, then
$$2a + 2b + 4ab = -1 \Rightarrow (2a + 1)(2b + 1) = 0,$$
which implies $a = -\frac{1}{2}$ or $b = -\frac{1}{2}$, impossible. Hence $a * b \in G$.
(ii) Associative:
$$\begin{aligned} a * (b * c) &= a + (b + c + 2bc) + 2a(b + c + 2bc) \\ &= a + b + c + 2ab + 2ac + 2bc + 4abc. \end{aligned}$$
Also,
$$\begin{aligned} (a * b) * c &= (a + b + 2ab) + c + 2(a + b + 2ab)c \\ &= a + b + c + 2ab + 2ac + 2bc + 4abc. \end{aligned}$$
Hence associativity holds.
(iii) Identity:
$$a * e = a \Rightarrow a + e + 2ae = a \Rightarrow e(1 + 2a) = 0.$$
Therefore
$$\boxed{e = 0}.$$
(iv) Inverse:
$$\begin{aligned} a * a^{-1} &= 0 \\ a + a^{-1} + 2aa^{-1} &= 0 \\ a^{-1}(1 + 2a) &= -a \end{aligned}$$
so
$$\boxed{a^{-1} = -\frac{a}{1 + 2a}}.$$
Since $a \neq -\frac{1}{2}$, this exists and belongs to G.
(v) Commutative:
$$a * b = a + b + 2ab = b + a + 2ba = b * a.$$
Hence $(G, *)$ is an abelian group.

2. **For** $G = \mathbb{R} - \{-1\}$ **and** $a * b = a + b + ab$**, show that** $(G, *)$ **is an abelian group.**
Answer: (i) Closure: If $a, b \neq -1$,
$$a * b = a + b + ab.$$
If $a * b = -1$, then
$$a + b + ab + 1 = 0 \Rightarrow (a + 1)(b + 1) = 0,$$
which is impossible. Therefore $a * b \in G$.
(ii) Associative:
$$\begin{aligned} a * (b * c) &= a + (b + c + bc) + a(b + c + bc) \\ &= a + b + c + ab + ac + bc + abc. \end{aligned}$$

Also,

$$\begin{aligned}(a*b)*c &= (a+b+ab)+c+(a+b+ab)c\\ &= a+b+c+ab+ac+bc+abc.\end{aligned}$$

Hence associativity holds.

(iii) Identity:

$$a*e=a \Rightarrow a+e+ae=a \Rightarrow e(1+a)=0.$$

Thus

$$\boxed{e=0}.$$

(iv) Inverse:

$$a*a^{-1}=0$$

$$a+a^{-1}+aa^{-1}=0$$

$$a^{-1}(1+a)=-a$$

so

$$\boxed{a^{-1}=-\frac{a}{1+a}}.$$

(v) Commutative:

$$a*b=a+b+ab=b+a+ba=b*a.$$

Therefore $(G,*)$ is an abelian group.

3. **Prove that the order of a subgroup of a finite group divides the order of the group.**
Answer: This is Lagrange's theorem.
Let G be a finite group of order n and H be a subgroup of order m. Let the distinct right cosets of H in G be

$$Ha_1, Ha_2, \ldots, Ha_r.$$

If

$$H=\{h_1,h_2,\ldots,h_m\},$$

then

$$Ha_i=\{h_1a_i,h_2a_i,\ldots,h_ma_i\}.$$

The mapping $h \mapsto ha_i$ is one-to-one, hence every right coset contains exactly m elements. Distinct cosets are disjoint and all cosets together cover G. Therefore

$$|G|=r|H|.$$

That is,

$$n=rm.$$

Hence

$$\boxed{|H| \text{ divides } |G|}.$$

4. **Show that $\mathbb{Q}^+$ forms an abelian group under $a*b=\dfrac{ab}{2}$.**
Answer: (i) Closure: For $a,b\in\mathbb{Q}^+$,

$$a*b=\frac{ab}{2}\in\mathbb{Q}^+.$$

(ii) Associative:

$$a*(b*c)=a*\frac{bc}{2}=\frac{abc}{4}.$$

Also,

$$(a*b)*c=\frac{ab}{2}*c=\frac{abc}{4}.$$

Hence associative.
(iii) Identity: Let e satisfy $a * e = a$.

$$\frac{ae}{2} = a \Rightarrow e = 2.$$

Thus $e = 2 \in \mathbb{Q}^+$.
(iv) Inverse: Let a^{-1} satisfy

$$a * a^{-1} = 2.$$

Then

$$\frac{aa^{-1}}{2} = 2 \Rightarrow aa^{-1} = 4,$$

so

$$\boxed{a^{-1} = \frac{4}{a}}.$$

(v) Commutative:

$$a * b = \frac{ab}{2} = \frac{ba}{2} = b * a.$$

Hence $(\mathbb{Q}^+, *)$ is an abelian group.

5. **Prove that every subgroup of a cyclic group is cyclic.**
Answer: Let

$$G = \langle a \rangle$$

and let $H \leq G$. If $H = \{e\}$, then H is cyclic. Suppose $H \neq \{e\}$.
Since every element of G is a power of a, there exists a least positive integer m such that

$$a^m \in H.$$

We claim

$$H = \langle a^m \rangle.$$

Certainly every power of a^m belongs to H, so

$$\langle a^m \rangle \subseteq H.$$

Let $a^k \in H$. By the division algorithm,

$$k = mq + r, \qquad 0 \leq r < m.$$

Then

$$a^r = a^{k-mq} = a^k (a^m)^{-q}.$$

Since $a^k \in H$ and $a^m \in H$, we get $a^r \in H$. By minimality of m, $r = 0$. Hence $k = mq$, so

$$a^k = (a^m)^q \in \langle a^m \rangle.$$

Therefore $H \subseteq \langle a^m \rangle$. Hence

$$\boxed{H = \langle a^m \rangle},$$

so H is cyclic.

6. **Prove that any two right cosets of H in G are either disjoint or identical.**
Answer: Let Ha and Hb be two right cosets. Suppose they are not disjoint. Then there exists x such that

$$x \in Ha \cap Hb.$$

Hence for some $h_1, h_2 \in H$,

$$x = h_1 a = h_2 b.$$

Therefore

$$a = h_1^{-1}h_2 b.$$

Since $h_1^{-1}h_2 \in H$, write $h_1^{-1}h_2 = h$. Then

$$a = hb.$$

Now

$$Ha = H(hb) = (Hh)b = Hb.$$

Thus if two right cosets intersect, they are identical. Therefore any two right cosets are either disjoint or identical. The proof for left cosets is similar.

7. **Prove that the intersection of two subgroups is a subgroup. Give an example to show the union need not be a subgroup.**
Answer: Let $H, K \leq G$. Since $e \in H$ and $e \in K$,

$$e \in H \cap K.$$

Let $a, b \in H \cap K$. Then

$$a, b \in H \quad \text{and} \quad a, b \in K.$$

Since H and K are subgroups,

$$ab^{-1} \in H \quad \text{and} \quad ab^{-1} \in K.$$

Hence

$$ab^{-1} \in H \cap K.$$

Therefore $H \cap K$ is a subgroup.
For the union, take the group $(\mathbb{Z}, +)$ with

$$H = 2\mathbb{Z}, \qquad K = 3\mathbb{Z}.$$

Then $2 \in H \subseteq H \cup K$ and $3 \in K \subseteq H \cup K$, but

$$2 + 3 = 5 \notin H \cup K.$$

Hence $H \cup K$ need not be a subgroup.

8. **Let M_2 be the set of all nonsingular 2×2 real matrices. Prove that M_2 is a group under usual matrix multiplication. Is it abelian?**
Answer: Let

$$M_2 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : ad - bc \neq 0 \right\}.$$

(i) Closure: If $A, B \in M_2$, then

$$\det(AB) = \det(A)\det(B) \neq 0.$$

Hence $AB \in M_2$.
(ii) Associative: Ordinary matrix multiplication is associative:

$$(AB)C = A(BC).$$

(iii) Identity:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in M_2,$$

and $AI = IA = A$.
(iv) Inverse: For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad ad - bc \neq 0,$$

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \in M_2.$$

Thus M_2 is a group.
It is **not abelian**. For example,

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix},$$

so $AB \neq BA$.

9. **Prove that $H \leq G$ is normal iff $a^{-1} * h * a \in H$ for every $a \in G$ and $h \in H$.**
Answer: ($\Rightarrow$) Assume H is normal. Then

$$a^{-1}Ha = H$$

for every $a \in G$. Hence for every $h \in H$,

$$a^{-1} * h * a \in H.$$

($\Leftarrow$) Assume

$$a^{-1} * h * a \in H$$

for every $a \in G$ and $h \in H$. Then

$$a^{-1}Ha \subseteq H.$$

Replace a by a^{-1}. We get

$$aHa^{-1} \subseteq H.$$

Conjugating by a^{-1} gives

$$H \subseteq a^{-1}Ha.$$

Therefore

$$a^{-1}Ha = H.$$

Hence H is normal in G.

10. **Prove that the order of each subgroup of a finite group is a divisor of the order of the group.**
Answer: Let G be finite and $H \leq G$. Let the distinct right cosets be

$$Ha_1, Ha_2, \ldots, Ha_r.$$

Each coset has $|H|$ elements because $h \mapsto ha_i$ is a one-to-one correspondence from H to Ha_i. Distinct cosets are disjoint and together form G. Hence

$$|G| = r|H|.$$

Therefore

$$\boxed{|H| \mid |G|}.$$

This is Lagrange's theorem.